

Notes: Efing, Goldbach & Nitsch (RFS, 2023)

Freeze! Financial Sanctions and Bank Responses

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1 Big picture

- **Question.** How do financial sanctions change bank lending when banks can lend through affiliates located in different legal and enforcement environments?
- **Core mechanism.** Sanctions raise litigation and compliance costs for banks doing business in targeted countries. Those costs differ across locations because enforcement intensity and anticrime institutions differ.
- **Main finding.** Banks in Germany reduce positions in sanctioned countries, while foreign affiliates of German banking groups increase positions relative to banks at home.
- **Formal enactment versus enforcement.** Affiliates increase lending not only when their host countries do not enact the sanctions, but also when they formally enact sanctions yet lack strong enforcement capacity.
- **Economic takeaway.** Sanctions do not simply reduce financial flows; they can redirect flows toward locations where compliance and enforcement costs are lower.

2 Incremental contribution

- The paper moves the sanctions literature from aggregate cross-border flows to the organizational geography of multinational banks.
- It separates formal legal enactment from actual enforcement capacity.
- It provides micro evidence on nonfinancial risk-taking by banks: litigation and compliance risk rather than credit or market risk.
- It extends regulatory-arbitrage logic to sanctions and financial-crime enforcement.
- It shows that subsidiaries and branches respond differently, linking legal form to sanctions enforcement.

3 Data and measurement

The core data source is the External Position Report of the Deutsche Bundesbank from March 2002 to December 2015. The data contain German banks' assets and liabilities with non-German counterparties, disaggregated by destination country and by reporting bank location. The main dependent variable is the log of one plus all external assets of bank b with counterparties in country c in month t .

The sanctions data identify financial sanctions imposed by Germany on behalf of the EU or UN. The Global Sanctions Database identifies whether the host countries of German bank affiliates also enact the same sanctions. FATF membership proxies for stronger anti-financial-crime institutions and enforcement capacity.

Interpretation: The data allow the authors to study not just whether German banking groups cut lending after sanctions, but where inside the multinational banking group the remaining activity is located.

4 Models and empirical specifications

4.1 Baseline difference-in-differences model

$$\log assets_{bct} = \beta EU/UN\ sanctions_{ct} + \alpha_{bc} + \alpha_{bt} + \varepsilon_{bct}.$$

Interpretation: Bank-counterparty-country fixed effects absorb permanent relationships; bank-time fixed effects absorb bank-level shocks.

4.2 Location heterogeneity

$$\log assets_{bct} = \beta_1 Sanction_{ct} + \beta_2 Sanction_{ct} \times Abroad_b + \alpha_{bc} + \alpha_{bt} + \varepsilon_{bct}.$$

Interpretation: β_1 measures banks in Germany; β_2 measures the relative response of foreign affiliates.

4.3 Formal enactment versus enforcement

$$\log assets_{bct} = \beta_1 Sanction_{ct} + \beta_2 Sanction_{ct} \times Enacts_{bct} + \beta_3 Sanction_{ct} \times DoesNotEnact_{bct} + \alpha_{bc} + \alpha_{bt} + \varepsilon_{bct}.$$

Interpretation: The specification separates host countries that formally enact sanctions from those that do not, and later splits by FATF enforcement proxies.

5 Identification and empirical design

The core problem is that sanctions are not randomly imposed. Country-time fixed effects absorb political, macroeconomic, and credit-demand shocks in the sanctioned country. Bank-time fixed effects absorb shocks to each bank. Banking-group by counterparty-country by time fixed effects compare affiliates within the same German banking group lending to the same country in the same month.

6 Section-by-section study guide

- Institutional setting: explains EU/UN sanctions, branches, subsidiaries, and compliance costs.
- Hypotheses: location matters; formal enactment is insufficient; subsidiaries should respond more than branches.
- Data: describes Bundesbank positions, sanctions episodes, FATF proxies, and branch/subsidiary legal form.
- Results: Table 4 and Figure 2 show the baseline reallocation; Tables 5–6 show formal enactment versus enforcement; Table 7 shows subsidiaries versus branches; Tables 8–9 rule out alternatives.

7 Tables and figures

7.1 Table 1: List of financial sanctions newly imposed between 2002 and 2015

Read: Lists 13 sanctioned counterparty countries, the announcement date, additional nonfinancial sanctions, and whether the sanction is EU-only or UN-wide.

Detailed interpretation: Defines the event set used in the empirical design and shows that the treatment is based on specific country-month sanction episodes.

Economic message: These rare events generate the identifying variation for bank-location reallocation.

Table 1
List of financial sanctions newly imposed between 2002 and 2015

	Announcement (1)	Additional nonfinancial sanctions (2)	Imposed by (3)
Iraq	Dec. 9, 1992	Restrictions of contracts and business activities	Entire UN
Myanmar	May 22, 2000	Export restriction on military equipment	Only EU
Liberia	Sept. 4, 2003 (lifted Jun. 20, 2016)	Export restriction on military equipment	Entire UN
Lebanon	Feb. 21, 2006	–	Entire UN
Belarus	May 18, 2006	Export restriction on military equipment	Only EU
Iran	Feb. 2, 2007	Export restriction on military equipment, chemicals, and other resources	Entire UN
Tunisia	Feb. 4, 2011	–	Only EU
Libya	Mar. 2, 2011	Export restriction on military equipment	Entire UN
Egypt	Mar. 21, 2011	–	Only EU
Syria	May 9, 2011	Export restriction on military equipment, chemicals, and other resources	Only EU
Afghanistan	Aug. 1, 2011	–	Entire UN
Russia	Mar. 5, 2014	Export restriction on oil drilling machinery, chemicals, and other resources	Only EU
Yemen	Dec. 18, 2014	–	Entire UN

This table lists the 13 countries in our sample that are sanctioned in at least 1 month between 2002 and 2015 and in which at least one German bank owns a position. *Sources:* Deutsche Bundesbank, Service center “Financial Sanctions,” and authors’ own investigations.

Table 1: List of financial sanctions newly imposed between 2002 and 2015

7.2 Table 2: Position-level summary statistics

Read: Panel A describes the full sample; Panel B compares sanctioned and never-sanctioned counterparty countries.

Detailed interpretation: German bank affiliates abroad account for nearly half of the sample. Sanctioned countries are much smaller and more volatile than never-sanctioned countries, making fixed effects essential.

Economic message: The sample is large enough to study reallocation, but treated exposures are small relative to total German foreign banking activity.

Table 2
Position-level summary statistics

A. Full sample

	Obs. (1)	Mean (2)	SD (3)	Skew. (4)
Host country indicators of German bank:				
<i>Abroad_b</i>	2,390,051	0.49	0.50	
<i>Outside EU_b</i>	2,390,051	0.18	0.39	
<i>Outside FATF_b</i>	2,390,051	0.08	0.26	
<i>In counterp. country_b</i>	2,390,051	0.02	0.15	
Sanction indicators:				
<i>EU/UN sanctions_{c,t}</i>	2,390,051	0.028	0.166	
<i>Bank host country enacts</i>	2,390,051	0.025	0.157	
<i>EU/UN sanctions_{b,c,t}</i>				
External assets <i>Assets_{b,c,t}</i> by counterparty (in EUR 1,000):				
All counterparty types	2,390,051	231,440.79	2,510,851.22	44.86
Banks	2,390,051	96,146.07	1,308,037.50	47.59
Other financials	2,390,051	9,056.44	301,561.90	96.76
Government	2,390,051	3,434.71	66,409.87	49.34
Nonfinancial firms, households, nonprofit	2,390,051	61,873.03	1,181,520.67	68.74
External assets <i>Assets_{b,c,t}</i> by asset class (in EUR 1,000):				
Loans	2,390,051	170,510.24	2,129,962.98	45.63
Nonloan assets	2,390,051	60,930.55	567,338.02	40.74
Treasury bills	2,390,051	698.19	30,171.74	106.06
Money market funds	2,390,051	2,796.53	84,405.42	11.81
Fixed-income securities	2,390,051	45,446.67	352,945.25	29.06
Shares	2,390,051	5,683.31	247,760.77	95.95
External assets <i>Assets_{b,c,t}</i> by host country (in EUR 1,000):				
Banks inside DEU	1,220,100	234,620.10	1,962,711	34.17
Banks abroad	1,169,951	228,125.20	2,976,845	44.78
Banks outside EU	438,153	223,229.60	2,498,737	29.29
Banks outside FATF	180,446	195,182.80	2,828,796	31.97
<i>Log (total) assets_{b,c,t}</i>	2,386,762	5.92	4.85	0.13

(Continued)

Table 2
(Continued)

B. Sanctioned versus never sanctioned counterparty countries

	Sanctioned at least once			Never sanctioned		
	Obs. (1)	Mean (2)	SD (3)	Obs. (4)	Mean (5)	SD (6)
Host country indicators of German bank:						
<i>Abroad_b</i>	159,107	0.43	0.49	2,230,944	0.49	0.50
<i>Outside EU_b</i>	159,107	0.15	0.35	2,230,944	0.19	0.39
<i>Outside FATF_b</i>	159,107	0.07	0.26	2,230,944	0.08	0.26
<i>In counterp. country_b</i>	159,107	0.01	0.07	2,230,944	0.02	0.15
External assets <i>Assets_{b,c,t}</i> by counterparty (in EUR 1,000):						
All counterparty types	159,107	37,118.39	268,686.54	2,230,944	245,299.52	2,597,298.05
Banks	159,107	14,480.36	148,578.98	2,230,944	101,970.32	1,353,109.47
Other financials	159,107	112.92	2,938.54	2,230,944	9,694.27	312,119.37
Government	159,107	1,139.94	24,179.01	2,230,944	3,598.37	68,430.31
Nonfinancial firms, households, nonprofit	159,107	17,995.81	112,783.02	2,230,944	65,002.27	1,222,496.02
External assets <i>Assets_{b,c,t}</i> by asset class (in EUR 1,000):						
Loans	159,107	33,729.03	218,300.31	2,230,944	180,265.23	2,203,512.38
Nonloan assets, i.e.,	159,107	3,389.36	84,019.39	2,230,944	65,034.28	586,575.99
Treasury bills	159,107	96.32	6,050.41	2,230,944	741.11	31,186.84
Fixed-income securities	159,107	1,879.82	52,433.59	2,230,944	48,553.78	364,847.09
Shares	159,107	895.83	35,198.47	2,230,944	6,024.74	256,267.81
External assets <i>Assets_{b,c,t}</i> by host country (in EUR 1,000):						
Banks inside DEU	90,819	41,081.21	231,612	1,129,281	250,184.80	2,038,252
Banks abroad	68,288	31,848.07	311,149	1,101,663	240,291.70	3,066,328
Banks outside EU	23,096	24,355.77	163,979	415,057	234,294.90	2,566,573
Banks outside FATF	11,228	4,837.94	19,427	169,218	207,811.60	2,920,695
<i>Log (total) assets_{b,c,t}</i>	158,908	3.69	4.29	2,227,854	6.08	4.85

This table provides summary statistics for the external positions in 105 foreign countries of 192 domestic banks in

(a) Panel A: Full sample

(b) Panel B: Sanctioned versus never sanctioned

7.3 Figure 1: External positions of German banks + Table 3: Country-level summary statistics

Read: Figure 1 plots external positions by bank location; Table 3 summarizes country-level assets and growth.

Detailed interpretation: Aggregate exposure to sanctioned countries falls, but the key empirical question is whether the remaining activity shifts across bank locations.

Economic message: Sanctions can reduce total exposure while redirecting residual exposure toward affiliates abroad.

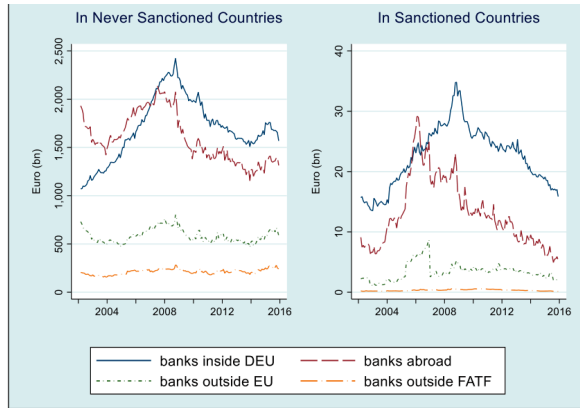


Figure 1
External positions of German banks This figure shows the evolution of total external positions owned by German banks with counterparties in countries that have never been sanctioned (left-hand graph) or have been sanctioned at least once between 2002 and 2015 (right-hand graph). Positions are aggregated over all counterparties, asset classes, and currencies of denomination. They are broken down into total external assets of all banks located inside Germany (DEU), of all German bank affiliates abroad, and of the subset of German bank affiliates that are located outside the EU or outside the FATF. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations.

(a) Figure 1

Table 3
Country-level summary statistics

	Obs. (1)	Mean (2)	SD (3)	Skew. (4)	P10 (5)	P50 (6)	P90 (7)
<i>A. All counterparty countries</i>							
Aggregated Assets _{c,t} (EUR bn)	17,008	32.50	100.00	5.68	0.12	3.27	66.70
Asset growth rate _{c,t}	16,851	0.159	6.743	72.28	-0.101	0.000	0.111
Volatility (asset growth rate) _c	105	1.072	7.514	9.64	0.045	0.098	0.431
<i>B. Never-sanctioned countries</i>							
Aggregated Assets _{c,t} (EUR bn)	14,850	36.90	107.00	5.30	0.27	4.05	93.60
Asset growth rate _{c,t}	14,715	0.147	7.060	71.56	-0.093	0.000	0.103
Volatility (asset growth rate) _c	92	0.965	7.917	9.43	0.048	0.091	0.270
<i>C. Sanctioned countries</i>							
Aggregated Assets _{c,t} (EUR bn)							
All periods	2,158	2.74	7.29	3.60	0.00	0.18	5.13
Before imposition of sanctions	1,021	4.29	9.98	2.43	0.00	0.15	22.60
After imposition of sanctions	1,137	1.34	2.67	4.20	0.00	0.38	4.00
Asset growth rate _{c,t}	2,136	0.243	3.920	23.36	-0.201	0.000	0.246
Volatility (asset growth rate) _c							
All assets	13	1.833	3.646	1.91	0.045	0.369	9.332
Only loans	13	1.938	3.620	1.87	0.043	0.499	9.332
Only nonloan assets	13	36.071	55.387	1.22	0.323	4.324	146.47

This table reports summary statistics for aggregated bank assets (in EUR bn) in 105 counterparty countries. One observation is a counterparty country-month pair (c, t). We further report the monthly growth rate of aggregated bank assets and the volatility of asset growth calculated over 2002 and 2015 at the country level. Panel A shows summary statistics for the full sample of all 105 counterparty countries. Panels B and C show separate statistics for the 92 counterparty countries that are never sanctioned between 2002 and 2015 and for the 13 countries that are sanctioned at least once during our sample period. Panel C further reports separate statistics for aggregated bank positions before and after the imposition of financial sanctions. Aggregated positions are calculated based on a random extraction of Deutsche Bundesbank's External Position Report that covers 90% of all external positions of all German banks worldwide between March 2002 and December 2015. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations.

(b) Table 3

7.4 Table 4: The effect of financial sanctions on lending across bank locations + Figure 2

Read: Table 4 shows that banks in Germany contract, while foreign affiliates increase relative to domestic banks. Figure 2 shows the event-study divergence after sanctions.

Detailed interpretation: The interaction with *Abroad* is positive and significant, and the effect survives counterparty-country-time fixed effects and within-group tests.

Economic message: Sanctions change the geography of German bank credit.

7.5 Table 5 and Table 6: Formal enactment versus enforcement

Read: Table 5 uses subsample regressions; Table 6 uses pooled regressions with rich fixed effects.

Detailed interpretation: Affiliates in non-enacting host countries expand the most, but affiliates in outside-FATF host countries also expand even when those countries formally enact sanctions.

Table 4: The effect of financial sanctions on lending across bank locations

	Dependent variable: $\log assets_{b,c,t}$									
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
$EUEN_{sanctions_{b,c,t}}$	-0.275*	-0.410***			-0.117*			-0.281*		
	(0.164)	(0.096)			(0.181)			(0.161)		
$EUEN_{sanctions_{b,c,t}} \times Abroad_b$		0.152***	0.111***	0.761***						
		(0.127)	(0.100)	(0.230)						
$EUEN_{sanctions_{b,c,t}} \times Outside FATF_b$					0.830**	0.730**	0.340**			
					(0.372)	(0.321)	(0.171)			
$EUEN_{sanctions_{b,c,t}} \times FATF_blacklist_b$								1.104**	1.111**	0.708
								(0.461)	(0.460)	(0.469)
Bank $\times$ counterpart country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Bank $\times$ time FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Counterparty country $\times$ time FE	No	No	Yes	No	No	Yes	No	Yes	No	Yes
Banking group $\times$ country $\times$ time FE	No	No	No	No	No	No	No	No	No	No
Obs.	2,377,900	2,377,900	2,377,900	1,439,900	2,377,900	1,439,900	2,377,900	2,377,900	1,439,900	
R ²	.840	.841	.844	.878	.840	.844	.878	.833	.858	.878

This table shows coefficient estimates from panel regressions. The dependent variable $\log assets_{b,c,t}$ is the natural logarithm of one plus all external assets of bank b with counterparties in country c in month t . The regressor $EUEN_{sanctions_{b,c,t}}$ is an indicator variable that equals one if Germany is imposing a financial sanction on behalf of the EU or UN on country c in month t (and zero otherwise). The regressors $Abroad_b$, $Outside FATF_b$, and $FATF_blacklist_b$ are indicator variables that equal one if bank b is located respectively outside Germany, outside the FATF, or in a country that has been declared noncooperative by the FATF (and zero otherwise). Specifications control for bank $\times$ counterpart country, bank $\times$ time, counterpart country $\times$ time, and banking group $\times$ counterpart country $\times$ time fixed effects as indicated. The sample contains bank-counterparty country-month observations for years 2002 to 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterpart country, and by month. Sources: Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, external position of banks (AUSTR), March 2002 to December 2015, and authors' own calculations. * $p < .1$, ** $p < .05$, *** $p < .01$.

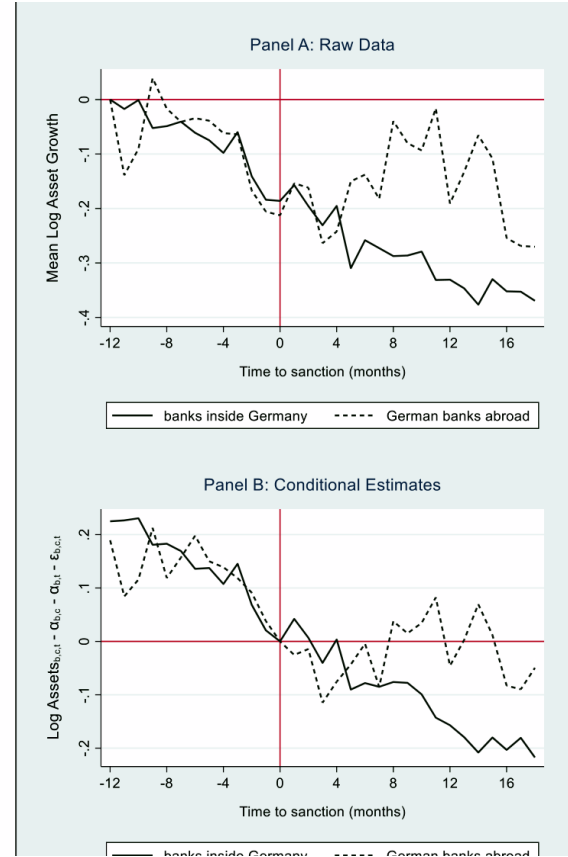


Figure 2: Banks' external positions around the introduction of financial sanctions

Economic message: Formal legal adoption is not enough; enforcement capacity drives economic effects.

Table 5
Formal enactment versus enforcement of financial sanctions: Subsample regressions

Dependent variable: $\text{Log assets}_{b,c,t}$	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Regression sample:							
Bank host country:	Germany	Foreign	Foreign	Inside FATF	Inside FATF	Outside FATF	Outside FATF
Host country enacts EU/UN sanctions:	Yes	Yes	No	Yes	No	Yes	No
$\text{EU/UN sanctions}_{c,t}$	-0.291* (0.144)	-0.032 (0.184)	0.725* (0.339)	-0.236** (0.096)	0.783* (0.435)	0.460** (0.177)	0.426*** (0.099)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	90,759	42,466	25,610	119,011	28,611	5,549	5,663
No. of banking groups	163	39	51	144	126	9	19
No. of individual banks	163	155	243	266	289	31	67
R^2	.85	.75	.83	.82	.85	.79	.75
Coefficient comparison:	Diff	SE	t-stat	Confidence interval			
Column 1 vs. column 6	0.751	(0.265)	2.84	[0.174; 1.328]			
Column 4 vs. column 6	0.696	(0.303)	2.30	[0.036; 1.357]			

This table shows coefficient estimates from panel regressions with bank $\times$ counterparty country fixed effects. The variables $\text{Log assets}_{b,c,t}$ and $\text{EU/UN sanctions}_{c,t}$ are defined as in Table 4. In columns 1 to 3, we split our data set into three subsamples based on banks' host countries: Germany, foreign bank host countries that enact EU/UN sanctions themselves, and foreign bank host countries that do not enact EU/UN sanctions like Germany. In columns 4 to 7, the data set is split into four subsamples depending on whether bank host countries are inside or outside the FATF and whether they enact or do not enact EU/UN sanctions like Germany. In all columns, the regression samples only include observations with counterparty countries against which Germany imposes EU/UN sanctions at least once between 2002 and 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterparty country, and by month in columns 2 to 7 and by counterparty country and month in column 1. To calculate the differences between the coefficients in columns 1 and 4 and column 6 and their statistical significance, we pool the relevant subsamples and estimate regressions with an interaction term between $\text{EU/UN sanctions}_{c,t}$ and a dummy variable for the subsample in column 6. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 5: Subsample regressions

Table 6
Formal enactment versus enforcement of financial sanctions: Pooled regressions

Dependent variable: $\text{Log assets}_{b,c,t}$	(1)	(2)	(3)	(4)
$\text{EU/UN sanctions}_{c,t}$	-0.470*** (0.068)		-0.363*** (0.135)	
$\text{EU/UN sanctions}_{c,t} \times \text{Bank host country is foreign and enacts EU/UN sanctions}_{b,c,t}$	0.444*** (0.123)	0.432*** (0.155)		
$\text{EU/UN sanctions}_{c,t} \times \text{Bank host country is foreign and does not enact EU/UN sanctions}_{b,c,t}$	0.846*** (0.253)	0.803** (0.306)		
$\text{EU/UN sanctions}_{c,t} \times \text{Bank host country is inside the FATF and does not enact EU/UN sanctions}_{b,c,t}$			0.664*** (0.294)	0.627* (0.336)
$\text{EU/UN sanctions}_{c,t} \times \text{Bank host country is outside the FATF and enacts EU/UN sanctions}_{b,c,t}$			0.915** (0.375)	0.792** (0.329)
$\text{EU/UN sanctions}_{c,t} \times \text{Bank host country is outside the FATF and does not enact EU/UN sanctions}_{b,c,t}$			0.801** (0.378)	0.737** (0.330)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes
Bank $\times$ time FE	Yes	Yes	Yes	Yes
Counterparty country $\times$ time FE	No	Yes	No	Yes
Obs.	2,377,900	2,377,900	2,377,900	2,377,900
R^2	.840	.844	.840	.844

This table shows coefficient estimates from panel regressions. The variables $\text{Log assets}_{b,c,t}$ and $\text{EU/UN sanctions}_{c,t}$ are defined as in Table 4. In columns 1 and 2, the indicator variable *Bank host country is foreign and enacts EU/UN sanctions* $_{b,c,t}$ (respectively, *Bank host country is foreign and does not enact EU/UN sanctions* $_{b,c,t}$) equals one for bank locations abroad that have enacted (respectively, have not enacted) the EU/UN sanctions against counterparty country c by time t (and zero otherwise). Similarly, in columns 3 and 4, the indicator variables *Bank host country is inside the FATF and does not enact EU/UN sanctions* $_{b,c,t}$, *Bank host country is outside the FATF and enacts EU/UN sanctions* $_{b,c,t}$, and *Bank host country is outside the FATF and does not enact EU/UN sanctions* $_{b,c,t}$ distinguish between bank locations inside and outside the FATF that either have or have not enacted the EU/UN sanctions against counterparty country c by time t (and zero otherwise). Specifications control for bank $\times$ counterparty country, bank $\times$ time, and counterparty country $\times$ time fixed effects as indicated. The sample contains bank-counterparty country-month observations for years 2002 to 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterparty country, and by month. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 6: Pooled regressions

7.6 Table 7: Branches versus subsidiaries

Read: Subsidiaries show stronger positive responses than branches.

Detailed interpretation: Subsidiaries are legally more independent and more tied to host-country enforcement than branches, so they are a more natural vehicle for sanctions-related reallocation.

Economic message: Legal form inside multinational banks matters for sanctions compliance.

Table 7
The effect of financial sanctions on bank branches versus subsidiaries

Dependent variable: $\text{Log assets}_{b,c,t}$	(1)	(2)	(3)	(4)	(5)
Regression sample:	German bank subsidiaries abroad		German bank branches abroad		
$EU/UN \text{ sanctions}_{c,t}$	-0.021 (0.015)	-0.141** (0.058)	-0.018 (0.411)	-0.037 (0.438)	-0.037 (0.438)
$EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country does not enact } EU/UN \text{ sanctions}_{b,c,t}$	0.542*** (0.187)		0.332 (0.364)		
$EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is inside the FATF and does not enact } EU/UN \text{ sanctions}_{b,c,t}$		0.652*** (0.221)		0.355 (0.391)	0.355 (0.391)
$EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is outside the FATF and enacts } EU/UN \text{ sanctions}_{b,c,t}$		0.909* (0.503)		confid'l	
$EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is outside the FATF and does not enact } EU/UN \text{ sanctions}_{b,c,t}$		0.761* (0.438)		confid'l	
$EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is outside the FATF}_b$					0.225 (0.438)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes	Yes
Bank $\times$ time FE	Yes	Yes	Yes	Yes	Yes
Obs.	562,864	562,864	597,325	597,325	597,325
R^2	.827	.827	.810	.810	.810

This table shows coefficient estimates from panel regressions. In columns 1 and 2, we restrict the regression sample to German bank subsidiaries located abroad. In columns 3, 4, and 5, the sample comprises only German bank branches abroad. The dependent variable $\text{Log assets}_{b,c,t}$ and the regressors are defined as in Tables 4 and 6. In column 4, data confidentiality requirements prevent us from showing the coefficient estimates of $EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is outside the FATF and enacts } EU/UN \text{ sanctions}_{b,c,t}$ and $EU/UN \text{ sanctions}_{c,t} \times \text{Bank host country is outside the FATF and does not enact } EU/UN \text{ sanctions}_{b,c,t}$ because each of them individually is identified by the branches of less than five German banking groups. All specifications control for bank $\times$ counterparty country and bank $\times$ time fixed effects. The sample contains bank-counterparty country-month observations for the years 2002 to 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterparty country, and by month. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 7: Branches versus subsidiaries

7.7 Tables 8 and 9: Alternative explanations

Read: Table 8 controls for cultural, linguistic, and geographic proximity. Table 9 controls for financial regulation and ex ante lending behavior.

Detailed interpretation: The outside-FATF result remains robust, indicating that enforcement quality is not merely a proxy for proximity, weak general regulation, or cyclical lending patterns.

Economic message: The mechanism is sanctions enforcement, not just relationship proximity or general regulation.

8 Detailed result map

- Hypothesis 1: supported by Table 4 and Figure 2.
- Hypothesis 2: supported by Tables 5 and 6.
- Hypothesis 3: supported by Table 7.
- Alternative explanations: addressed by Tables 8 and 9.

9 Short summary

- Banks in Germany reduce positions after sanctions.
- Foreign affiliates increase relative to banks at home.
- The increase is strongest in weak-enforcement host countries.

Table 8
Controlling for cultural, linguistic, and geographic proximity

Dependent variable: $\log assets_{b,c,t}$	(1)	(2)	(3)	(4)
$EU/UN\ sanctions_{c,t}$	-0.354** (0.151)	-0.462** (0.187)	-0.373** (0.175)	-0.317* (0.160)
$EU/UN\ sanctions_{c,t} \times Outside\ FATF_b$	0.839** (0.362)	0.799** (0.362)	0.826** (0.368)	0.803** (0.393)
$EU/UN\ sanctions_{c,t} \times Same\ official\ language_{b,c}$	0.677** (0.323)			
$EU/UN\ sanctions_{c,t} \times Spoken\ language\ overlap_{b,c}$		1.389*** (0.407)		
$EU/UN\ sanctions_{c,t} \times Native\ language\ overlap_{b,c}$			4.121*** (1.299)	
$EU/UN\ sanctions_{c,t} \times In\ neighbor\ country_{b,c}$				0.283 (0.497)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes
Bank $\times$ time FE	Yes	Yes	Yes	Yes
Obs.	2,312,126	2,262,616	2,262,616	2,377,900
R^2	.840	.835	.835	.840

This table shows coefficient estimates from panel regressions. $\log assets_{b,c,t}$, $EU/UN\ sanctions_{c,t}$, and $Outside\ FATF_b$ are defined as in Table 4. $Same\ official\ language_{b,c}$ is an indicator variable that equals one if bank b is located in a host country that has the same official language as counterparty country c . $Spoken\ language\ overlap_{b,c}$ and $Native\ language\ overlap_{b,c}$ measure the probabilities that two people from bank b 's host country and from counterparty country c speak at least one common language or, respectively, share the same native tongue (Melitz and Toubal 2014). $In\ neighbor\ country_{b,c}$ is an indicator variable that equals one if the host country of bank b and counterparty country c share a border. Specifications control for bank $\times$ counterparty country and bank $\times$ time fixed effects. The sample contains bank-counterparty country-month observations for the years 2002 to 2015. Robust standard errors (reported in parentheses) are clustered by bank host country, counterparty country, and by month. *Sources:* Research Data and Service Centre (RDSC) of the Deutsche Bundesbank, the external position of banks (AUSTA), from March 2002 to December 2015, and the authors' own calculations. * $p < .1$; ** $p < .05$; *** $p < .01$.

$sanctions_{c,t} \times Outside\ FATF_b$ remains significant and its magnitude barely

Table 8: Proximity controls

Table 9
Controlling for the intensity of financial regulation and for banks' ex ante lending behavior

Dependent variable: $\log assets_{b,c,t}$	(1)	(2)	(3)	(4)	(5)
$EU/UN\ sanctions_{c,t}$	-0.320** (0.161)	-0.343** (0.146)	-0.314* (0.162)	-0.163 (0.161)	-0.544** (0.245)
$EU/UN\ sanctions_{c,t} \times Outside\ FATF_b$	0.840** (0.373)	0.694** (0.331)	0.849** (0.373)	1.040*** (0.338)	1.001** (0.438)
Controlling for the intensity of financial regulation:					
$EU/UN\ sanctions_{c,t} \times Official\ supervisory\ power_{b,t} < P25$		0.338* (0.200)			
$EU/UN\ sanctions_{c,t} \times Capital\ regulatory\ index_{b,t} < P25$			0.639 (0.396)		
$EU/UN\ sanctions_{c,t} \times External\ governance\ index_{b,t} < P25$				-0.137 (0.101)	
Controlling for banks' ex ante lending behavior outside sanction episodes:					
$EU/UN\ sanctions_{c,t} \times Lending\ sensitivity_b$ (Agg. lending, Agg. borrower-GDP)				0.065 (0.245)	
$EU/UN\ sanctions_{c,t} \times Lending\ sensitivity_b$ (Agg. lending, Home GDP of b)					-0.578** (0.245)
$EU/UN\ sanctions_{c,t} \times Lending\ sensitivity_{b,c}$ (Lending in c , GDP of c)					1.687*** (0.180)
$EU/UN\ sanctions_{c,t} \times Lending\ sensitivity_{b,c}$ (Lending in c , Home GDP of b)					-0.423 (0.283)
Bank $\times$ counterparty country FE	Yes	Yes	Yes	Yes	Yes
Bank $\times$ time FE	Yes	Yes	Yes	Yes	Yes
Obs.	2,377,900	2,377,900	2,377,900	2,251,318	1,507,272
R^2	.840	.840	.840	.845	.822

This table shows coefficient estimates from panel regressions. $\log assets_{b,c,t}$, $EU/UN\ sanctions_{c,t}$, and $Outside\ FATF_b$ are defined as in Table 4. In columns 1 to 3, we interact $EU/UN\ sanctions_{c,t}$ with three binary variables that are equal to one for bank locations with weak *Official supervisory power*, *Capital regulatory index*, and *External governance index*, below the 25th percentile (and zero otherwise). The three variables are indexes that Barth, Caprio, and Levine (2013) calculate based on the World Bank surveys on bank regulation. In columns 4 and 5, we interact $EU/UN\ sanctions_{c,t}$ with different proxies *Lending sensitivity* for the ex ante cyclicality of bank lending outside sanctions episodes. *Lending sensitivity* (Aggregate lending, Aggregate borrower-GDP) is calculated as the correlation between bank b 's total positions in non-sanctioned-host countries and these countries' aggregate GDP. *Lending sensitivity* (Aggregate lending, Home GDP of b) is the correlation between bank b 's total positions in non-sanctioned host countries and the GDP of bank b 's own host country. *Lending*

Table 9: Regulation and ex ante behavior controls

- Formal sanction enactment does not ensure uniform behavior if enforcement differs.
- Subsidiaries respond more strongly than branches.
- The results survive rich fixed effects and alternative-explanation tests.

10 Conclusion

10.1 Core conclusion

Financial sanctions can shift bank activity toward lax enforcement jurisdictions instead of uniformly reducing financial activity.

10.2 Economic intuition

Sanctions raise expected litigation and compliance costs. Banks in strict jurisdictions withdraw; affiliates in lax jurisdictions face lower expected costs and can gain business.

10.3 Incremental contribution revisited

The paper shows regulatory arbitrage in nonfinancial risk-taking and separates formal enactment from actual enforcement.

10.4 Policy implications

Rules must be enforced consistently across jurisdictions. Monitoring subsidiaries and weak-enforcement locations is central for sanctions effectiveness.

10.5 Limitations

The data do not observe individual counterparties, internal compliance costs, or illegal transactions directly; the paper studies German banking groups rather than all global banks.

10.6 Takeaway

Effective financial sanctions require harmonized enforcement capacity, not only harmonized legal rules.